RODRIGO GOMES DE ARAÚJO

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Education

Master's — **Sep 2025 → Present**
Computer Science — *University of Porto (FCUP)*

Bachelor's — **Sep 2022 → Jul 2025**
Computer Science and Engineering — *University of Porto (FEUP)*

Research

CaST (Cryptographically-aware Session Types) | *Master's Thesis* — **Jul 2026 → Present**

- Extending session types to capture **cryptographic resource usage**, so that misuse in protocols is rejected at compile-time.
- Formalizing the extension with a **soundness proof**, integrating it into the FreeST compiler in **Haskell**.
- Built a **Kademlia DHT in FreeST** and the **syntax highlighting extension** for Zed.

Experience

Murph | *Co-founder & Lead Engineer* — **Jul 2026 → Present**

- Co-founded and lead a **3-person startup**. Primary author of the codebase and owner of the architecture.
- Murph merges **version control, project management and CI** into one system behind a simple **VSCode extension**.
- **Typestate** guarantees no change reaches main without **verification** and **human review**.
- **Signed approvals** bound to the exact action, **tiered authority** for agents and humans and a **tamper-evident history**.
- **Git underlay** with **compare-and-swap** writes, so verification and agents run **in parallel** in isolated worktrees without overwriting main.
- Light runtime with **crash-safe persistence**. State is rebuilt from Git against an atomic checkpoint, caching only what each computation needs.

Coursework

PoW Blockchain for Decentralized Auctions | *Data Systems Security*, 19/20 — *Rust, Tokio, libp2p*

- **Research on the state of the art** of Bitcoin, Ethereum and Solana, Kademlia and churn/attack literature before building.
- **Kademlia DHT** overlay for decentralized network construction, **GossipSub** block propagation, **chain synchronization and fork resolution**.
- PoW chain used as a public ledger for an **auction client**. Blake2b hashing, Ed25519 signatures and Merkle Trees for transaction validation.
- Docker **adversarial network simulation** with bootstrap, well behaved and **malicious nodes**.

Distributed Protocols | *Distributed Systems*, 19/20 — *Rust, Tokio, Tonic*

- **Totally ordered multicast** with **Lamport clocks**, defended against timestamp reuse, backdating and far-future attacks.
- **Token ring mutual exclusion** with **failure detection** and automatic ring healing.
- **Push-pull gossip** with exponential **anti-entropy** for state convergence.

Portuguese Elections Data Platform | *Advanced Topics in Databases*, 18/20 — *PostgreSQL, Python, OCaml*

- **ETL pipeline** normalizing **8 official elections** from **inconsistent** government spreadsheets.
- **Automated reconciliation** against official totals (votes, turnout, seats) and **coherence checks**.
- Triggers for data ingestion and **D'Hondt** seats recomputed in SQL as a cross-check.
- **Star-schema warehouse** with **cube/rollup**, **window functions** and **recursive drill-down** to 3,000+ parishes.
- Vote-swing analytics and **PostGIS maps** served by an **OCaml** API to a **Bonsai** UI.

Achievements and Awards

Top 10 of 130 Startups, Fall 2026 Academia Cohort with Murph | Unicorn Factory Lisboa — **Oct 2026**

2nd Place, IEEE RetroJam (48-hour game in Rust) | IEEE UP Student Branch — **Oct 2025**

Participation in RoboCup 2016 | RoboCup Federation — **Jun 2016**

1st Place in the National Robotics Championship | Instituto Politécnico de Bragança — **May 2016**

Technical Skills

Languages: Rust, Java, Python, C/C++, SQL, Haskell
Distributed Systems & Cryptography: Tokio, libp2p, gRPC, Ed25519, Blake2b, Merkle Trees
Data & Infrastructure: PostgreSQL, pandas, NumPy, scikit-learn, FastAPI, Docker, Terraform, Git, UNIX/Linux
Web: React, Dioxus, Bonsai, Dream, HTML5, CSS
Spoken Languages: Portuguese (Native), English (Fluent), Spanish (Intermediate)